

Detailed Explanation of Wind Averaging Periods and Spectral Bins

1. Wind Averaging Periods

The different averaging periods represent how wind speed is measured over different time intervals. This is crucial for engineering design because structures respond differently to wind over different durations.

10-Minute Average Wind (Reference)

Aspect	Explanation
What it is	The standard meteorological wind speed averaged over 10 minutes
Why 10 minutes?	This is the WMO (World Meteorological Organization) standard for reporting wind speeds
What ERA5 provides	ERA5 gives 10-minute average winds (this is your reference)
Engineering use	Baseline for all calculations, building codes reference
Gust factor	$\times 1.00$ (reference)

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Example: If 10-min avg = 10 m/s
This means over 10 minutes, the average wind speed was 10 m/s

3-Minute Average Wind

Aspect	Explanation
What it is	Average wind speed over a 3-minute period
Why 3 minutes?	Captures shorter-term fluctuations that 10-min averages miss
How calculated	10-min avg $\times 1.15$ (empirical factor for over-water)
Engineering use	Dynamic response of structures (small buildings, towers)

Aspect	Explanation
Gust factor	×1.15

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Example: 10-min avg = 10 m/s
3-min avg = $10 \times 1.15 = 11.5$ m/s

10-Second Gust

Aspect	Explanation
What it is	Peak wind speed averaged over 10 seconds
Why 10 seconds?	Captures gust fronts and short-term peak events
How calculated	10-min avg × 1.35 (empirical factor for over-water)
Engineering use	Mooring line peak tensions, floating structure displacement
Gust factor	×1.35

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Example: 10-min avg = 10 m/s
10-sec gust = $10 \times 1.35 = 13.5$ m/s

3-Second Gust (Peak Gust)

Aspect	Explanation
What it is	Peak instantaneous wind speed over 3 seconds
Why 3 seconds?	Standard for ultimate limit state design (ASCE 7, Eurocode)
How calculated	10-min avg × 1.45 (empirical factor for over-water)
Engineering use	Ultimate loads, structural design, panel uplift forces
Gust factor	×1.45

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Example: 10-min avg = 10 m/s
3-sec gust = $10 \times 1.45 = 14.5$ m/s

Why Different Factors for Over-Water vs Over-Land?

Surface	3-sec Gust Factor	Reason
Over-water	1.45	Smoother surface, less friction, higher gusts
Over-land	1.50-1.65	Rougher surface, more turbulence
Lake Kariba	1.45 (over-water)	Your location is on the lake

Summary Table

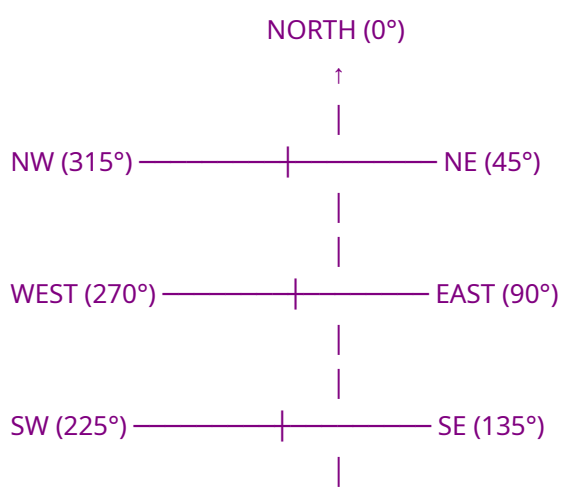
Period	Factor	Application
10-min mean	×1.00	Reference, building codes, wave modeling input
3-min mean	×1.15	Dynamic response, small vessel operations
10-sec gust	×1.35	Mooring loads, floating structure dynamics
3-sec gust	×1.45	Ultimate limit state, panel uplift, structural design

2. Spectral Directional Bins (10° resolution)

What are Spectral Directional Bins?

In wave modeling, waves come from different directions. SWAN divides the full circle (360°) into discrete "bins" or sectors to represent wave energy coming from each direction.

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SOUTH (180°)

36 Directional Bins (10° each)

Parameter	Value
Total directions	36 bins
Bin width	10° ($360^\circ \div 36 = 10^\circ$)
What it means	Wave energy is calculated for each 10° sector

Visual Representation of 10° Bins

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Full circle (360°) divided into 36 bins of 10° each:

Bin 1: 0° - 10° (N to N10°E)

Bin 2: 10° - 20°

Bin 3: 20° - 30°

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Bin 9: 80° - 90° (East)

...

Bin 18: 170° - 180° (South)

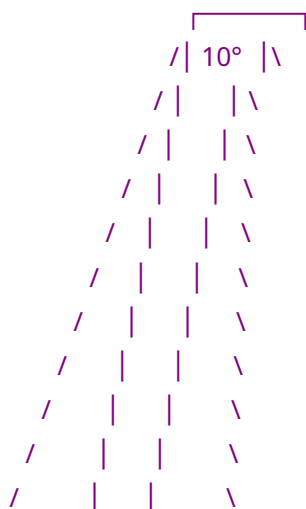
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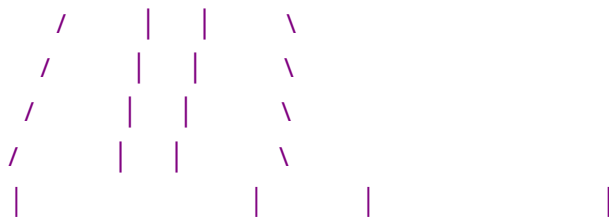
Bin 27: 260° - 270° (West)

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Bin 36: 350° - 360° (N10°W to N)

10° bins shown as wedges:





Why 36 Bins (10° resolution)?

Resolution	Number of Bins	Accuracy	Computation Time
5°	72 bins	Higher	2× slower
10°	36 bins	Good	Standard
15°	24 bins	Lower	Faster

10° resolution is the SWAN default because:

- Wind sea typically spreads over 30-60°
- 10° captures this spreading well
- Good balance between accuracy and computational speed
- Recommended for most coastal applications

What This Means for Your Wave Model

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If wind is from EAST (90°):

Bin containing 90° will have highest wave energy

Adjacent bins (80-90° and 90-100°) will have some energy

Bins far from 90° (like 0-10°) will have little or no energy

This creates a directional spread of wave energy

Engineering Significance

Aspect	Implication
Mooring design	Waves from multiple directions require 360° mooring
Shoreline impact	Wave direction affects erosion patterns
Floating solar	Panels should be oriented perpendicular to dominant wave

Aspect	Implication
	direction
Fetch length	Direction determines how far wind travels over water

In Your SWAN Configuration

swan

CGRID REGULAR ... CIRCLE 36 0.04 1.00 25

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This "36" means 36 directional bins (10° resolution)

Summary: Factors and Bins in Your Tool

Feature	Value	Meaning
10-min avg	×1.00	Reference wind speed from ERA5
3-min avg	×1.15	Shorter-term average
10-sec gust	×1.35	Peak over 10 seconds
3-sec gust	×1.45	Ultimate peak gust
Spectral bins	36 bins	10° directional resolution
Frequency bins	25 bins	Logarithmic frequency distribution

These factors and bins ensure your wave model captures:

1. The correct peak wind loads for engineering design
2. The directional distribution of wave energy
3. Accurate wave height predictions for floating solar infrastructure